

Lecture Notes: Complex Numbers and Polar Coordinates

Calculus I

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Section 1 Complex Numbers

Definition 1 (Imaginary Unit) The imaginary unit, denoted by i , is defined by the property

$$i^2 = -1.$$

Equivalently, $i = \sqrt{-1}$.

Definition 2 (Complex Number) A complex number is a number of the form

$$z = a + bi,$$

where a and b are real numbers. The real number a is called the real part of z , denoted $\operatorname{Re}(z) = a$. The real number b is called the imaginary part of z , denoted $\operatorname{Im}(z) = b$. The set of all complex numbers is denoted by $\mathbb{C}$.

Remark 1 If $b = 0$, then $z = a$ is a real number, so $\mathbb{R} \subset \mathbb{C}$. If $a = 0$ and $b \neq 0$, then $z = bi$ is called a pure imaginary number.

Example 1 The following are complex numbers:

1. $z = 3 + 4i$, with $\operatorname{Re}(z) = 3$ and $\operatorname{Im}(z) = 4$.
2. $z = -2 - i$, with $\operatorname{Re}(z) = -2$ and $\operatorname{Im}(z) = -1$.
3. $z = 5$, with $\operatorname{Re}(z) = 5$ and $\operatorname{Im}(z) = 0$.
4. $z = 7i$, with $\operatorname{Re}(z) = 0$ and $\operatorname{Im}(z) = 7$.

Definition 3 (Equality of Complex Numbers) Two complex numbers $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$ are equal if and only if $a_1 = a_2$ and $b_1 = b_2$.

Definition 4 (Complex Conjugate) The complex conjugate of $z = a + bi$, denoted $\bar{z}$, is defined by

$$\bar{z} = a - bi.$$

Definition 5 (Modulus) The modulus (or absolute value) of $z = a + bi$, denoted $|z|$, is defined by

$$|z| = \sqrt{a^2 + b^2}.$$

Theorem 1 (Properties of the Conjugate and Modulus) For any complex numbers $z, z_1, z_2 \in \mathbb{C}$:

1. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$.
2. $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$.
3. $z\bar{z} = |z|^2$.
4. $|z_1 z_2| = |z_1| |z_2|$.
5. $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$, provided $z_2 \neq 0$.
6. $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle Inequality).

For (3): If $z = a + bi$, then $z\bar{z} = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 + b^2 = |z|^2$. The other properties follow by direct computation.

Section 2 Operations with Complex Numbers

Definition 6 (Addition and Subtraction) If $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$, then

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2)i,$$

$$z_1 - z_2 = (a_1 - a_2) + (b_1 - b_2)i.$$

Definition 7 (Multiplication) If $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$, then

$$z_1 z_2 = (a_1 + b_1i)(a_2 + b_2i) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i.$$

Definition 8 (Division) If $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i \neq 0$, then

$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2} = \frac{(a_1 + b_1i)(a_2 - b_2i)}{a_2^2 + b_2^2} = \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + \frac{a_2 b_1 - a_1 b_2}{a_2^2 + b_2^2}i.$$

Example 2 Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$, compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, and $\frac{z_1}{z_2}$.

$$z_1 + z_2 = (3 + 1) + (4 - 2)i = 4 + 2i.$$

$$z_1 - z_2 = (3 - 1) + (4 + 2)i = 2 + 6i.$$

$$z_1 z_2 = (3)(1) - (4)(-2) + [(3)(-2) + (1)(4)]i = 3 + 8 + (-6 + 4)i = 11 - 2i.$$

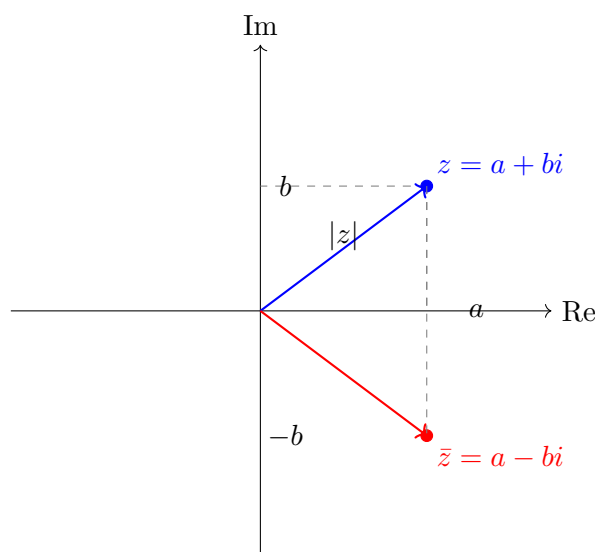
$$\frac{z_1}{z_2} = \frac{(3 + 4i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{3 + 6i + 4i + 8i^2}{1 + 4} = \frac{-5 + 10i}{5} = -1 + 2i.$$

Example 3 Compute i^n for various integers n .

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^5 = i, \dots$$

The powers of i cycle with period 4: $i, -1, -i, 1$.

Theorem 2 (Geometric Interpretation) A complex number $z = a + bi$ can be represented as the point (a, b) in the complex plane (or Argand diagram), where the horizontal axis is the real axis and the vertical axis is the imaginary axis. The modulus $|z|$ is the distance from the origin to the point (a, b) , and the conjugate $\bar{z}$ is the reflection across the real axis.



Section 3 Polar Coordinates

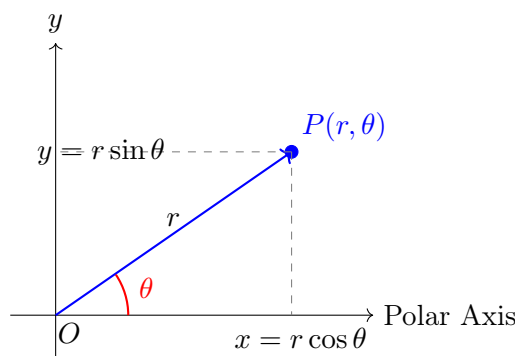
Definition 9 (Polar Coordinate System) The polar coordinate system represents points in the plane using an ordered pair (r, θ) , where r is the directed distance from the origin (called the pole), and θ is the angle measured counterclockwise from the positive x -axis (called the polar axis).

Theorem 3 (Conversion Between Cartesian and Polar Coordinates) Given Cartesian coordinates (x, y) and polar coordinates (r, θ) :

$$x = r \cos \theta, \quad y = r \sin \theta,$$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x} \quad (x \neq 0).$$

These follow directly from the definitions of sine and cosine in a right triangle with hypotenuse r and the Pythagorean theorem.



Example 4 Convert the polar point $(2, \frac{\pi}{6})$ to Cartesian coordinates.

$$x = 2 \cos \frac{\pi}{6} = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}, \quad y = 2 \sin \frac{\pi}{6} = 2 \cdot \frac{1}{2} = 1.$$

Thus, the Cartesian coordinates are $(\sqrt{3}, 1)$.

Example 5 Convert the Cartesian point $(-1, 1)$ to polar coordinates.

$$r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}.$$

Since the point is in the second quadrant, $\theta = \frac{3\pi}{4}$. Thus, $(\sqrt{2}, \frac{3\pi}{4})$.

Remark 2 (Non-Uniqueness of Polar Coordinates) A point in the plane can be represented by infinitely many polar coordinate pairs. If (r, θ) represents a point, then so do $(r, \theta + 2n\pi)$ for any integer n , and $(-r, \theta + (2n + 1)\pi)$ for any integer n .

Section 4 Graphing Polar Coordinate Equations

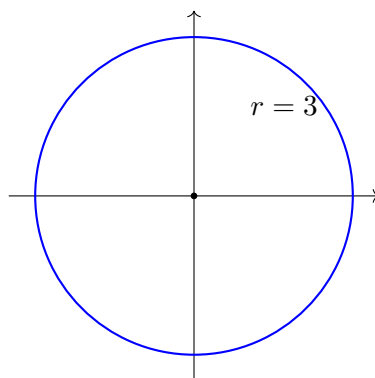
Definition 10 (Polar Curve) The graph of a polar equation $r = f(\theta)$ consists of all points P that have at least one polar representation (r, θ) satisfying the equation.

Theorem 4 (Symmetry Tests for Polar Graphs) 1. **Symmetry about the polar axis (x -axis):** The graph is symmetric if replacing (r, θ) with $(r, -\theta)$ or $(-r, \pi - \theta)$ yields an equivalent equation.

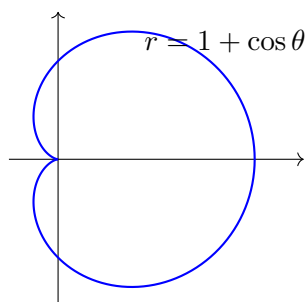
2. **Symmetry about the line $\theta = \frac{\pi}{2}$ (y-axis):** The graph is symmetric if replacing (r, θ) with $(r, \pi - \theta)$ or $(-r, -\theta)$ yields an equivalent equation.

3. **Symmetry about the pole (origin):** The graph is symmetric if replacing (r, θ) with $(-r, \theta)$ or $(r, \theta + \pi)$ yields an equivalent equation.

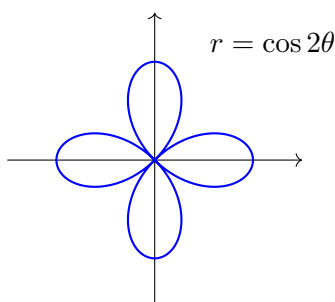
Example 6 (Circle) Sketch the graph of $r = 3$. This equation describes all points at a distance 3 from the pole. The graph is a circle of radius 3 centered at the origin.



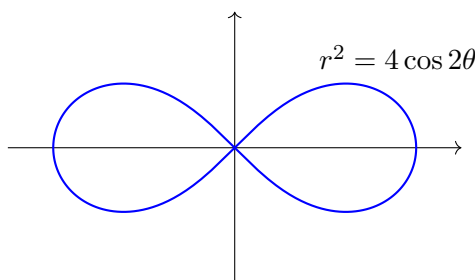
Example 7 (Cardioid) Sketch the graph of $r = 1 + \cos \theta$. This is a cardioid. Compute values: at $\theta = 0$, $r = 2$; at $\theta = \frac{\pi}{2}$, $r = 1$; at $\theta = \pi$, $r = 0$; at $\theta = \frac{3\pi}{2}$, $r = 1$. The graph is symmetric about the polar axis.



Example 8 (Rose Curve) Sketch the graph of $r = \cos 2\theta$. This is a four-petaled rose. At $\theta = 0$, $r = 1$; at $\theta = \frac{\pi}{4}$, $r = 0$; at $\theta = \frac{\pi}{2}$, $r = -1$; and so on. The curve has four petals.



Example 9 (Lemniscate) Sketch the graph of $r^2 = 4 \cos 2\theta$. This is a lemniscate (figure-eight shape). The curve is defined only when $\cos 2\theta \geq 0$, i.e., $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}] \cup [\frac{3\pi}{4}, \frac{5\pi}{4}]$.



Theorem 5 (Common Polar Curves) • **Circle:** $r = a$, or $r = 2a \cos \theta$, or $r = 2a \sin \theta$.

- **Cardioid:** $r = a \pm a \cos \theta$ or $r = a \pm a \sin \theta$.
- **Limaçon:** $r = a \pm b \cos \theta$ or $r = a \pm b \sin \theta$.
- **Rose:** $r = a \cos n\theta$ or $r = a \sin n\theta$ (has n petals if n is odd, $2n$ petals if n is even).
- **Lemniscate:** $r^2 = a^2 \cos 2\theta$ or $r^2 = a^2 \sin 2\theta$.
- **Spiral:** $r = a\theta$ (Archimedean) or $r = ae^{b\theta}$ (logarithmic).

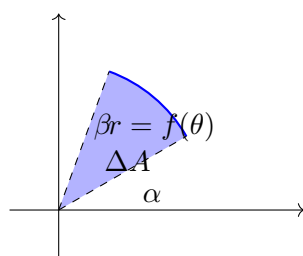
Section 5 Integration in Polar Coordinates

Definition 11 (Area Element in Polar Coordinates) A small sector of a polar curve with angle $d\theta$ and radius r has area approximately $\frac{1}{2}r^2 d\theta$. This is the basis for integration in polar coordinates.

Theorem 6 (Area in Polar Coordinates) If f is continuous and non-negative on the interval $[\alpha, \beta]$, then the area A of the region bounded by the graph of $r = f(\theta)$ and the rays $\theta = \alpha$ and $\theta = \beta$ is

$$A = \int_{\alpha}^{\beta} \frac{1}{2} [f(\theta)]^2 d\theta = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta.$$

Divide $[\alpha, \beta]$ into n subintervals with angles θ_i . In each subinterval, choose a sample angle θ_i^* and approximate the area by a circular sector with radius $f(\theta_i^*)$ and central angle $\Delta\theta$. The area of the sector is $\frac{1}{2}[f(\theta_i^*)]^2 \Delta\theta$. Summing and taking the limit as $n \rightarrow \infty$ gives the integral.



Example 10 Find the area enclosed by the cardioid $r = 1 + \cos \theta$. The curve is traced completely as θ goes from 0 to 2π .

$$A = \int_0^{2\pi} \frac{1}{2} (1 + \cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta.$$

Using $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$,

$$A = \frac{1}{2} \int_0^{2\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta = \frac{1}{2} \int_0^{2\pi} \left(\frac{3}{2} + 2 \cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta.$$

$$A = \frac{1}{2} \left[\frac{3\theta}{2} + 2 \sin \theta + \frac{\sin 2\theta}{4} \right]_0^{2\pi} = \frac{1}{2} \cdot 3\pi = \frac{3\pi}{2}.$$

Example 11 Find the area of one petal of the rose $r = \cos 2\theta$. One petal is traced as θ goes from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$.

$$A = \int_{-\pi/4}^{\pi/4} \frac{1}{2} \cos^2 2\theta \, d\theta = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \frac{1 + \cos 4\theta}{2} \, d\theta = \frac{1}{4} \left[\theta + \frac{\sin 4\theta}{4} \right]_{-\pi/4}^{\pi/4}.$$

$$A = \frac{1}{4} \left[\left(\frac{\pi}{4} + 0 \right) - \left(-\frac{\pi}{4} + 0 \right) \right] = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}.$$

Theorem 7 (Area Between Two Polar Curves) If $r_1 = f(\theta)$ and $r_2 = g(\theta)$ are continuous with $f(\theta) \geq g(\theta) \geq 0$ on $[\alpha, \beta]$, then the area of the region between them is

$$A = \int_{\alpha}^{\beta} \frac{1}{2} ([f(\theta)]^2 - [g(\theta)]^2) \, d\theta.$$

Example 12 Find the area inside $r = 2$ and outside $r = 1 + \cos \theta$. Intersection: $2 = 1 + \cos \theta \implies \cos \theta = 1 \implies \theta = 0$, but we need the region where $2 \geq 1 + \cos \theta$, i.e., $\cos \theta \leq 1$, which is true except at $\theta = 0$. The full area is

$$A = \int_0^{2\pi} \frac{1}{2} (4 - (1 + \cos \theta)^2) \, d\theta.$$

Since $(1 + \cos \theta)^2 = 1 + 2 \cos \theta + \cos^2 \theta$, we get

$$A = \frac{1}{2} \int_0^{2\pi} (3 - 2 \cos \theta - \cos^2 \theta) \, d\theta = \frac{1}{2} \int_0^{2\pi} \left(3 - 2 \cos \theta - \frac{1 + \cos 2\theta}{2} \right) \, d\theta.$$

$$A = \frac{1}{2} \int_0^{2\pi} \left(\frac{5}{2} - 2 \cos \theta - \frac{\cos 2\theta}{2} \right) \, d\theta = \frac{1}{2} \left[\frac{5\theta}{2} - 2 \sin \theta - \frac{\sin 2\theta}{4} \right]_0^{2\pi} = \frac{1}{2} \cdot 5\pi = \frac{5\pi}{2}.$$

Section 6 Arc Length in Polar Coordinates

Theorem 8 (Arc Length in Polar Coordinates) If f has a continuous derivative on $[\alpha, \beta]$, then the length L of the polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is

$$L = \int_{\alpha}^{\beta} \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} \, d\theta = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} \, d\theta.$$

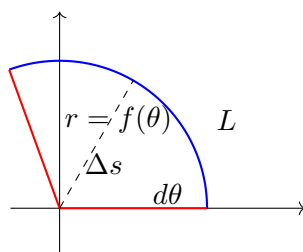
Parametrize the polar curve as $x = r \cos \theta = f(\theta) \cos \theta$ and $y = r \sin \theta = f(\theta) \sin \theta$. Then

$$\frac{dx}{d\theta} = f'(\theta) \cos \theta - f(\theta) \sin \theta, \quad \frac{dy}{d\theta} = f'(\theta) \sin \theta + f(\theta) \cos \theta.$$

Thus,

$$\left(\frac{dx}{d\theta} \right)^2 + \left(\frac{dy}{d\theta} \right)^2 = [f'(\theta)]^2 + [f(\theta)]^2 = r^2 + \left(\frac{dr}{d\theta} \right)^2.$$

The arc length formula for parametric curves gives the result.



Example 13 Find the length of the cardioid $r = 1 + \cos \theta$. Since the cardioid is symmetric about the polar axis, we can compute the length for $\theta \in [0, \pi]$ and double it. $r = 1 + \cos \theta$, $\frac{dr}{d\theta} = -\sin \theta$. Thus,

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = (1 + \cos \theta)^2 + \sin^2 \theta = 1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta = 2 + 2 \cos \theta = 2(1 + \cos \theta).$$

Using the identity $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$,

$$\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \sqrt{4 \cos^2 \frac{\theta}{2}} = 2 \left| \cos \frac{\theta}{2} \right| = 2 \cos \frac{\theta}{2} \quad \text{for } \theta \in [0, \pi].$$

Thus,

$$L = 2 \int_0^\pi 2 \cos \frac{\theta}{2} d\theta = 4 \left[2 \sin \frac{\theta}{2} \right]_0^\pi = 8(1 - 0) = 8.$$

Example 14 Find the length of the circle $r = 4 \sin \theta$. $r = 4 \sin \theta$, $\frac{dr}{d\theta} = 4 \cos \theta$. Then

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = 16 \sin^2 \theta + 16 \cos^2 \theta = 16.$$

So $\sqrt{r^2 + (dr/d\theta)^2} = 4$. The circle is traced completely as $\theta \in [0, \pi]$.

$$L = \int_0^\pi 4 d\theta = 4\pi.$$

This matches the circumference $2\pi r = 2\pi(2) = 4\pi$, since the circle has radius 2.

Example 15 Find the length of the spiral $r = e^\theta$ from $\theta = 0$ to $\theta = 2\pi$. $r = e^\theta$, $\frac{dr}{d\theta} = e^\theta$. Then

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = e^{2\theta} + e^{2\theta} = 2e^{2\theta}.$$

Thus,

$$L = \int_0^{2\pi} \sqrt{2e^{2\theta}} d\theta = \sqrt{2} \int_0^{2\pi} e^\theta d\theta = \sqrt{2} [e^\theta]_0^{2\pi} = \sqrt{2}(e^{2\pi} - 1).$$

Theorem 9 (Surface Area in Polar Coordinates) If f has a continuous derivative on $[\alpha, \beta]$ and $f(\theta) \geq 0$, the surface area S generated by revolving the polar curve $r = f(\theta)$ about the polar axis (x -axis) is

$$S = \int_\alpha^\beta 2\pi f(\theta) \sin \theta \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} d\theta.$$

If the curve is revolved about the line $\theta = \frac{\pi}{2}$ (y -axis), then

$$S = \int_\alpha^\beta 2\pi f(\theta) \cos \theta \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} d\theta.$$

Example 16 Find the surface area generated by revolving the cardioid $r = 1 + \cos \theta$ about the polar axis.

$$S = \int_0^\pi 2\pi(1 + \cos \theta) \sin \theta \cdot 2 \cos \frac{\theta}{2} d\theta.$$

Using $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ and $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$,

$$S = \int_0^\pi 8\pi \cos^3 \frac{\theta}{2} \sin \frac{\theta}{2} d\theta.$$

Let $u = \cos \frac{\theta}{2}$, $du = -\frac{1}{2} \sin \frac{\theta}{2} d\theta$. Limits: $\theta = 0 \rightarrow u = 1$, $\theta = \pi \rightarrow u = 0$.

$$S = 16\pi \int_0^1 u^3 du = 16\pi \cdot \frac{1}{4} = 4\pi.$$